Snubber-Based Suppression of Overshoot Caused by Three-Level Modulation in SiC MOSFET Full-Bridge Converters

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***Abstract*—Silicon carbide (SiC) MOSFETs are increasingly used in power electronics due to higher efficiency, power density, and faster switching compared to silicon (Si) devices. However, their fast switching can cause significant voltage spikes, especially at high frequencies. Adjusting gate resistance is often insufficient in three-level PWM modulation, where some MOSFETs remain exposed to overvoltage. This work investigates snubber circuits as an alternative. Simulations using PSIM integrated with LTspice show that snubbers effectively reduce voltage slope and overvoltage spikes, with the RCD snubber performing best, offering superior MOSFET protection compared to gate resistance adjustments.**

***Index Terms*—Three-Level PWM, Voltage Overshoot, Silicon Carbide (SiC) MOSFET, Snubber Circuits.**

I. Introduction

The growing demand for electricity in various applications has driven the development of increasingly efficient power electronic converters. In this context, as presented in [1], [2], the emergence of the silicon carbide (SiC) metal oxide-semiconductor field effect transistor (MOSFET) represents a true revolution. It turns out that silicon-based (Si) technologies have limitations in terms of efficiency, particularly in applications requiring high switching frequencies. SiC and GaN (Gallium Nitride) transistors are WBG semiconductors that offer several advantages over Si, as shown in Fig. 1.

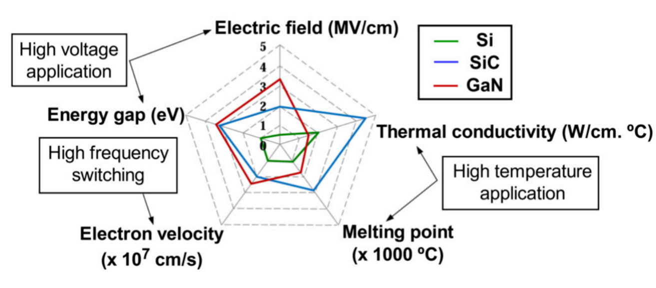

Fig. 1. Comparison of the material properties of Si and SiC.Adapted from [3].

These devices perform well in high frequency environments due to the lower intrinsic capacitance between the drain and source terminals (C_{DS}), the gate and drain (C_{GD}) and the gate and source (C_{GS}). These reduced capacitances allow SiC MOSFETs to switch on and off more quickly compared to Si transistors, resulting in more efficient operation, especially at higher frequencies [1].

However, the superior characteristics of SiC MOSFETs also introduce new challenges during high-frequency switching. During turn-off, the signal exhibits a voltage overshoot that exceeds (V_{DS}) and shows oscillations [4]. This phenomenon compromises switching quality and affects the overall efficiency of the system.

Although solutions to mitigate this effect are discussed in the literature, such as those presented in [5]– [6], including the adjustment of the gate resistance in the driving circuit, three-level modulation is limited to this application due to the switching sequence. Therefore, this paper presents SPICE simulations of a full-bridge converter based on SiC MOSFETs, highlighting another technique for overshoot reduction, with a focus on the use of snubber circuits. The simulation results are presented.

II. Voltage Spike Mitigation

A. Gate Resistor Adjustmen

For voltage overshoot issues, various research efforts have focused on techniques to mitigate in high-frequency applications of SiC MOSFETs. One of these simpler approaches consists of modifying the gate resistances, RG_{ON} and RG_{OFF}, shown in Fig. 2, which control, respectively, the turn-on and turn-off times, respectively, in the MOSFET drive circuit.

As illustrated in Fig. 3a and Fig. 3b, by increasing the gate resistance, a reduction in overvoltage peaks and oscillation frequency is observed during the MOSFET shutoff, which contributes to more stable operation with less oscillation. However, by increasing this gate resistance, as can be seen in Fig. 3b, the signal response experiences a delay. As discussed in [3], this effect leads to higher switching losses.Therefore, it becomes evident that this technique presents a direct trade-off between overvoltage, oscillations, and switching losses.

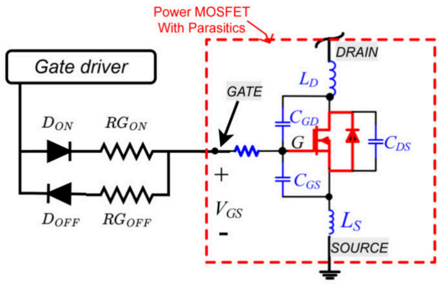

Fig. 2. External gate resistances of the SiC MOSFET driving circuit and its associated parasitic elements. Adapted from [4] respectively.

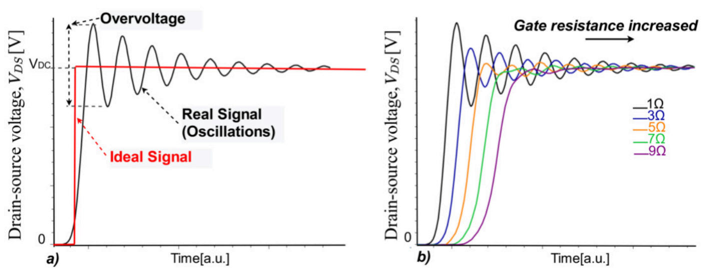

Fig. 3. (a) V_{DS} voltage of the MOSFET during turn-off; and (b) Influence of the gate resistance of the driving circuit on this voltage.

B. Snubber Circuits

Another technique for overvoltage reduction is the use of snubbers. Unlike increasing the gate resistance, which causes delays and higher switching losses, snubbers attenuate voltage spikes and oscillations during MOSFET turn-off without significantly affecting the switching time [7].

The simplest snubber commonly discussed in the literature is the C-type snubber, shown in Fig. 4a, whie 1r ch connects a capacitor C_{SA} in parallel with the MOSFETs of a power converter. The equivalent circuit during the turn-off period of Q_2 can be seen in Fig. 4b, composed of an inductance, L_P, associated with the layout traces and parasitic elements of the MOSFET, in series with the capacitor C_{SA} of the snubber.

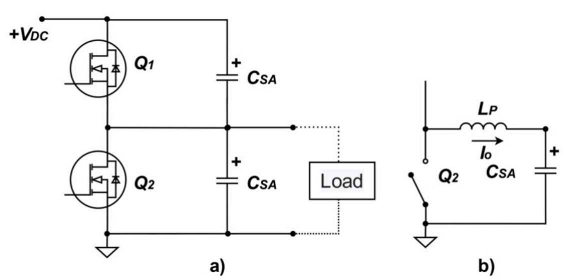

Fig. 4. (a) Half-bridge topology circuit with a C-type, and (b) equivalent circuit.

This capacitor increases the output capacitance of the MOSFET and switching energy during the turn-off of Q_2. As a result, voltage spikes can be attenuated, contributing to the protection of the MOSFET [8]. The expressions for C_{SA} and L_P are given by (1) and (2) [9]. I_{OFF} represents the current at the moment of MOSFET turn-off, similar to the drain-source current I_{DS} of Q_2 in the on-state. The V_{DSmax} voltage must be higher than the DC bus voltage (V_{DC}). L_P depends on the oscillation frequency f_{RING} and the parasitic capacitance C_P, which can be approximated by the output capacitance of the MOSFET C_{OSS} [10].

$$C_{SA} = \frac{L_P \cdot I_{OFF}^2}{V_{DS_{max}}^2} \tag{1}$$

$$L_P = \frac{1}{4 \cdot \pi^2 \cdot f_{RING}^2 \cdot C_P} \tag{2}$$

The same idea applies to the RC-type snubber, as shown in Fig. 5a, which includes a resistor R_{SB} in series with the capacitor C_{SB}, both connected in parallel with the MOSFETs Q_1 and Q_2. This resistor smooths the charge process of C_{SB} during the MOSFET turn-off, preventing unwanted resonances [9]. The equivalent circuit representing the behavior of MOSFET Q_2 during turn-off is shown in Fig. 5b, which forms a series RLC circuit.

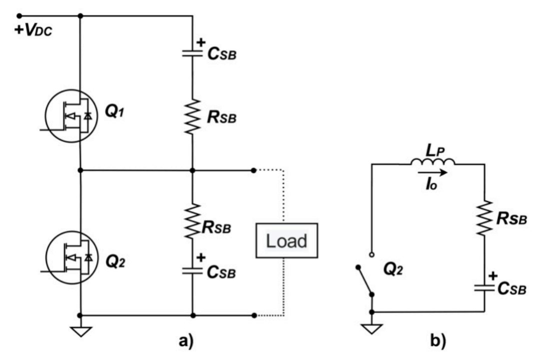

Fig. 5. (a) Half-bridge topology circuit with a RC-type, and (b) equivalent circuit.

Therefore, by using the RC type snubber, it is possible not only to reduce the voltage spikes in V_{DS} during MOSFET turn-off but also to attenuate the oscillation frequency [11]. This occurs because the resistor introduces an energy dissipation mechanism that dampens the system's oscillatory response. As a result, the RC snubber would contribute to more stable switching with lower voltage overshoot. The expressions for R_{SB} and C_{SB} are defined by (3) and (4), respectively [11]. It can be seen that the damping factor, ζ, appears in the expression for R_{SB}, which is usually set to 1 in RC snubbers, corresponding to the critical damping. Furthermore, the value of C_{SB} depends on R_{SB}.

$$R_{SB} = \frac{1}{2\zeta}\sqrt{\frac{L_P}{C_P}} \tag{3}$$

$$C_{SB} = \frac{1}{2\pi R_{SB} f_{RING}} \tag{4}$$

the RCD-type snubber, as shown in Fig. 6a, includes. On the other hand, the RCD-type snubber, as shown inFig. 6a,includes a resistor R_{SC}, a capacitor C_{SC}, and a diode D_S, enabling the clamping of the voltage through the diode. When Q_2 turns off Fig. 6b, the parasitic inductance energy is diverted to C_{SC} via D_S. During turn-on, R_{SC} dissipates this energy back to the bus.

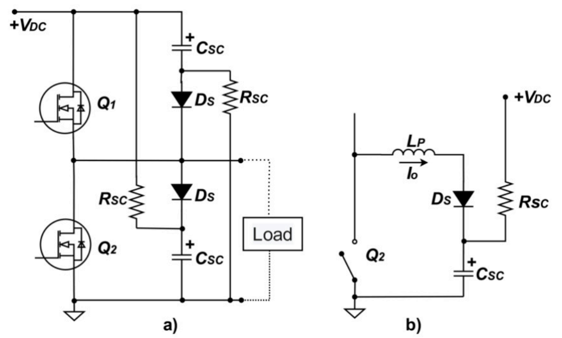

Fig. 6. (a) Half-bridge topology circuit with RCD-type, and (b) equivalent circuit.

This topology prevents MOSFET overvoltage, but D_S must be fast to avoid reverse recovery losses [10]. The capacitance C_{SC} is defined in (5) and depends on the voltage difference $V_{DS(p)} - V_{DC}$, with $V_{DS(p)}$ measured experimentally without the snubber. To ensure complete discharge before the next cycle, R_{SC} must satisfy the condition in (6), considering the switching frequency f_{SW} [10]-

$$C_{SC} = \frac{L_P \cdot I_{OFF}^2}{(V_{DS(p)} - V_{DC})^2} \tag{5}$$

$$R_{SC} \leq \frac{1}{2.3 \cdot C_{SC} \cdot f_{SW}} \tag{6}$$

III. Three-Level Modulation

A. Circuit Operation

The three-level pulse width modulation (PWM), used in the switching of MOSFETs in power converters such as the Full Bridge topology shown in Fig. 7, contributes to increasing the switching frequency seen by the grid, which reduces the size of passive filters [12]. However, this technique may impair voltage overshoot performance, since the switching of the MOSFETs is not directly controlled via the gate resistance [13].

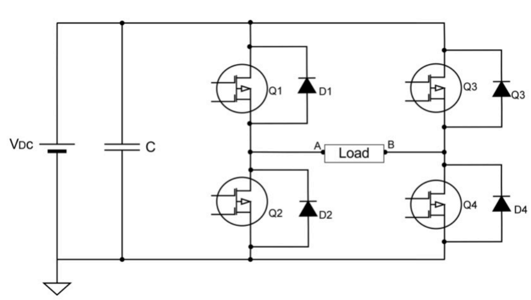

Fig. 7. Full-Bridge topology circuit with four MOSFET operation, where points A and B indicate the load connection.

The control signals for this modulation are illustrated in Fig. 8a. Unlike the two-level modulation, whose control signals and resulting load voltage are shown in Fig. 8c and Fig. 8d, respectively [14]. The three-level modulation introduces an intermediate 0 V level, as depicted in Fig. 8b [15]. In this case, the load voltage transitions in two steps: from +V_{DC} to 0 V and from 0 V to -V_{DC}.

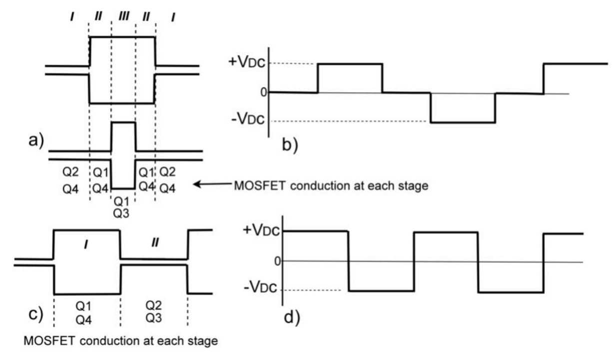

Fig. 8. (a) Control signals for three-level modulation; (b) Load signal for three-level modulation; (c) Control signals for two-level modulation; and (d) Load signal for two-level modulation.

However, to further reduce overvoltage effects in three-level modulation, changing the gate resistance in the drive circuit is no longer effective for MOSFET Q2 in Fig. 7. At the beginning of Region II in Fig. Fig. 8a, when turning off Q_2 , its drain-source current (I_{DS}) is not abruptly interrupted, as shown in Fig. 9a, since it is drained to diode D2, which controls conduction.

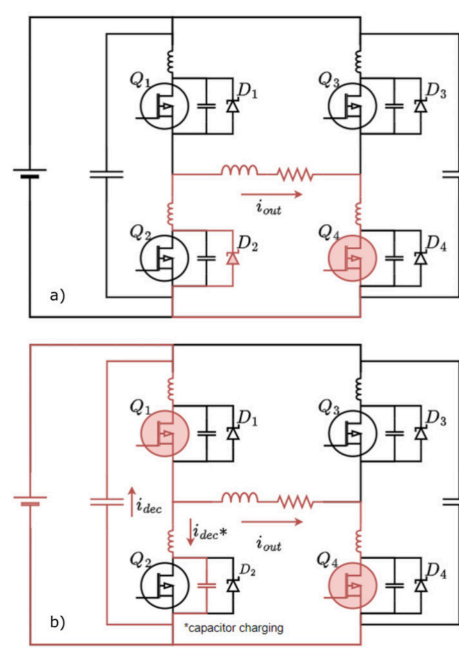

Fig. 9. (a) Transition between Sections I and II; (b) The portion corresponding to Section II of the three-level modulation.

Q_1 is turned on during this transient, shown in Fig. 9b, and soon conducts this same current. Thus, the voltage V_{DS}

of Q_2 increases dramatically, only due to the charging of the parasitic capacitance C_{DS}. In this scenario, increasing or decreasing the gate resistance of Q_2, which would typically be used to slow down the switching and control (dv/dt), is less effective because the turn-off process is dominated only by the turn-on process of Q_1. MOSFETs Q_1, Q_3, and Q_4, operate differently from Q_2. In this case, the adjustment of the gate resistance to control (dv/dt), and consequently mitigate voltage overshoot is valid. Thus, snubber circuits are an alternative to further reduce voltage spikes in the MOSFET Q_2. Snubbers are auxiliary circuits composed of resistors, capacitors, and diodes, designed to damp voltage spikes during switching transients [8]. These circuits are essential because, even with the use of high-speed capacitors, the overshoot can be reduced but not completely eliminated due to the parasitic inductances of both the capacitor and the circuit itself [13].

B. Validation of the spice model

To ensure accuracy in analyzing the high-frequency switching effects of the SiC MOSFET, the SPICE model of the C3M0021120K device, provided by Wolfspeed, was characterized [16]. This model was implemented in PSIM and integrated with LTspice through co-simulation. The SPICE model of the MOSFET was separated into parts, as shown in Fig. 10, allowing for more comprehensive analyses of the device behavior, as well as the possibility of modifying specific parameters.

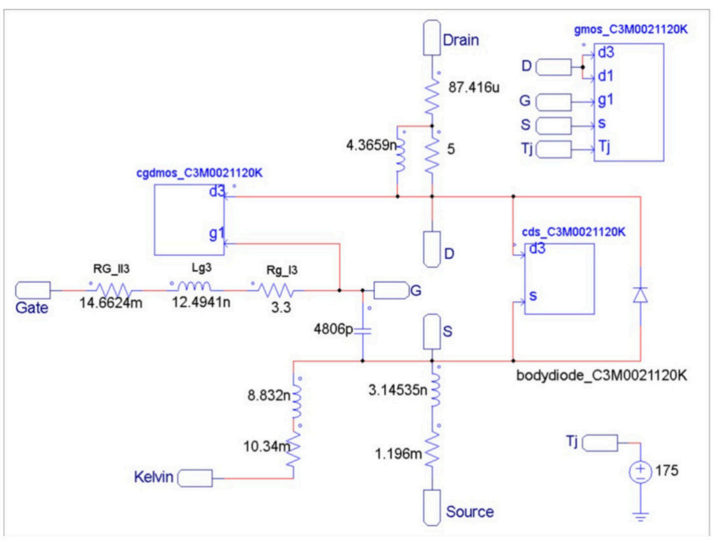

Fig. 10. SPICE model development of the SiC MOSFET C3M0021120K performed in PSIM software.

As part of the validation process, the output characteristic curves (I_{DS} versus V_{DS}) were simulated for different gate-source voltages (V_{GS}), following the same conditions as those in the manufacturer's datasheet. Fig. 11 presents these results, comparing the curve obtained from the simulation with the one provided in the datasheet at a junction temperature (Tj) of 25 °C. Additionally, validation was extended to Tj = 175 °C. As a result, by measuring the correlation coefficient between the datasheet and the simulated data, one can observe a 94% correlation, thus suggesting that the SPICE model, confirming that the SPICE model accurately reproduces the static behavior of the device. This validation is essential to ensure the reliability of the switching performance analyses in the subsequent stages of this study.

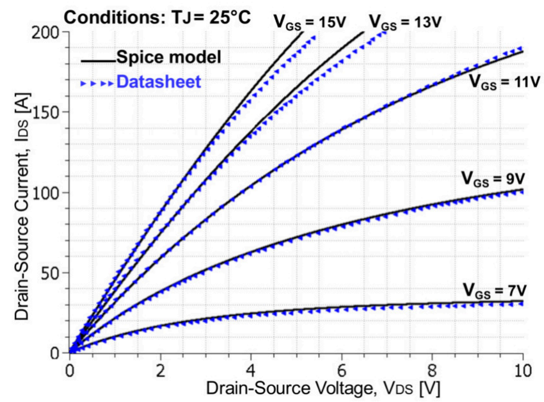

Fig. 11. Comparison of the $I_{\mathrm{DS}} \times V_{\mathrm{DS}}$ curve obtained from SPICE simulation and the one from the datasheet of the SiC MOSFET C3M0021120K, considering 25 °C.

IV. Simulation Results

For validation purposes, a simulation was performed using PSIM software integrated with LTSpice of a full-bridge converter, similar to the one shown in Fig. 7, operating with three-level modulation to drive MOSFETs Q_1, Q_2, Q_3, and Q_4. The SPICE model of the SiC MOSFET used was the C3M0021120K from Wolfspeed, with 1.2 kV technology [16]. It is important to note that no parameters of the SPICE model were altered.The switching frequency of the simulated circuit was set to 200 kHz. The gate drive voltage, V_{GS}, applied to the MOSFETs was bipolar, ranging from -3 V to +15 V. The V_{GS} signals for each MOSFET, following the order and symmetry of the three-level modulation, are shown in Fig. 12.

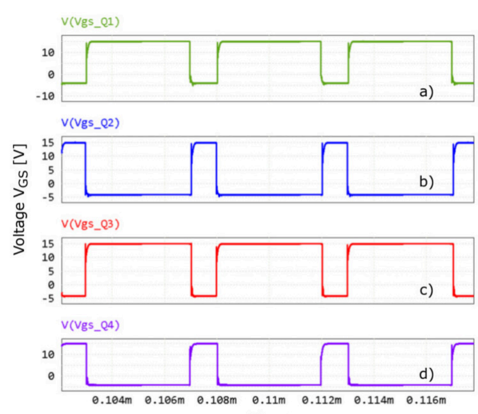

Fig. 12. Switching voltage (V_{GS}) signals of the four MOSFETs in the full-bridge topology for the three-level PWM: (a) Q1, (b) Q2, (c) Q3, and (d) Q4.

The converter load consisted of an RLC circuit connected between points A and B in Fig. 7. The bus voltage, V_{DC}, was set to 600 V. In the SPICE simulation, diodes D_1, D_2, D_3, and D_4 were added in parallel to MOSFETs Q_1, Q_2, Q_3, and Q_4, respectively, as the intrinsic diodes of the MOSFETs exhibit high losses and significant reverse recovery. The external diodes help to reduce these effects, even though the current splits between them and the intrinsic MOSFET diodes [12]. The model used for these diodes was the C4D20120A from Wolfspeed, as indicated in [16].

The designed values for each snubber are presented in Table I, calculated using (1) to (6). The parameters I_{OFF}, f_{RING}, L_P, and $V_{DS(p)}$ were obtained from the simulation, while C_P was extracted from the datasheet of the C3M0021120K MOSFET. The diode adopted in the RCD snubber was selected for its low reverse recovery characteristics, as is the case for the C4D40120D model [17]. Fig. 13a shows that the variation of gate resistance RG_{OFF} is not effective in reducing the overvoltage and oscillations in V_{DS} during MOSFET Q_2 switching under three-level modulation.

TABLE I
SNUBBER CONFIGURATIONS AND THEIR RESPECTIVE COMPONENTS.

Snubber	Capacitor (pF)	Resistor (Ω)	Diode Model
C	360	–	–
RC	360	8	–
RCD	354	10	C4D40120H

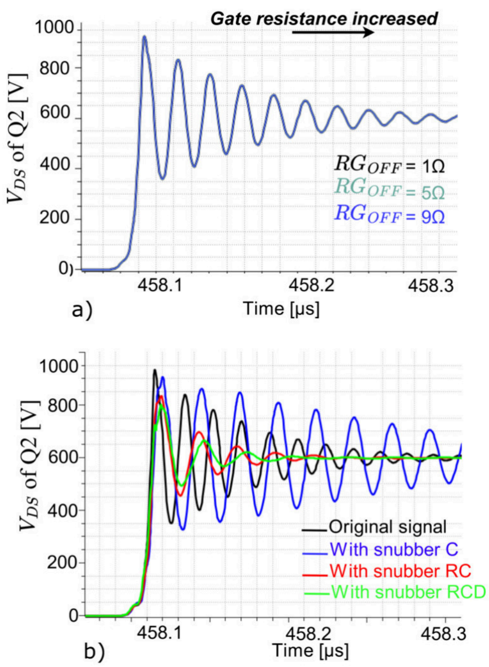

Fig. 13. V_{DS} voltage waveforms during turn-off events of Q_2 with varying $R_{G_{OFF}}$ resistance (a) and (b) snubber circuits.

The amplitude and oscillation pattern remain nearly unchanged. In Fig. 14a, during the turn-on of Q_1, increasing RG_{OFF} slightly reduces the overvoltage but introduces significant delays, potentially increasing switching losses [1]. After adding snubber circuits to Q_2 Fig. 13b, a significant reduction in overvoltage and oscillations is observed. The standalone capacitive snubber (type C) was less effective, while the RC and RCD types performed better, with the RCD being the most efficient. Similar results were found for Q_1 Fig. 14b. Even the C-type snubber had some effect, but the RCD snubber was also the most effective for MOSFETs Q_3 and Q_4.

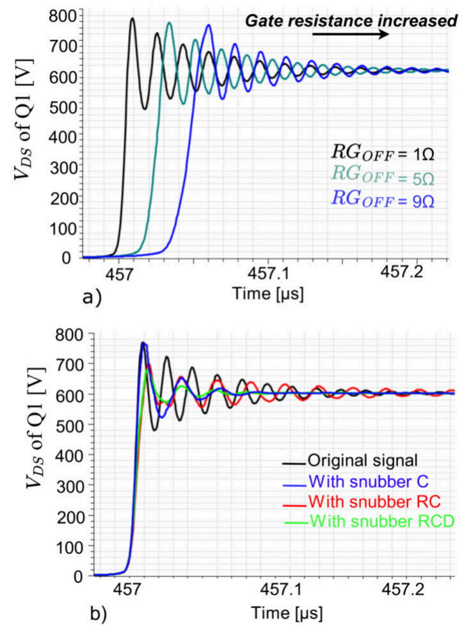

Fig. 14. V_{DS} voltage waveforms during turn-off events of Q_1 under (a) the same $R_{G_{OFF}}$ resistance and (b) snubber circuit conditions.

The results are summarized in Table II. One can observe that the application of different snubber techniques leads to a reduction in both the voltage slope and the voltage spikes magnitude. The original waveform exhibits the largest slope (35×10^9 V/s) and the highest peak voltage (925 V). With the addition of the C-type snubber, a slight attenuation occurs, thus reducing the slope to 30×10^9 V/s and the peak to 900 V. In addition, the RC-type snubber shows a more pronounced improvement, lowering the slope to 22×10^9 V/s and the peak to 840 V. The best performance was achieved with the RCD snubber, which produced the smallest slope (17×10^9 V/s) and lowest peak voltage (790 V), highlighting its superior effectiveness in damping oscillations and mitigating voltage spikes.

TABLE II
COMPARISON OF OVERVOLTAGE AND VOLTAGE SLOPE REDUCTION FOR DIFFERENT SNUBBER TECHNIQUES.

Technique	Slope ($\times 10^9$ V/s)	Voltage Spikes [V]
Original Signal	35	925
Snubber C	30	900
Snubber RC	22	840
Snubber RCD	17	790

Therefore, the aforementioned results suggest that snubber circuits are more effective in reducing the overvoltage in three-level modulation than a simple gate resistance adjustment. In this sense, snubbers are capable of preventing MOSFET overvoltage failures and improving devices reliability.

V. Conclusion

This work demonstrated that increasing the gate resistance is ineffective in reducing the voltage overshoot during Q_2 MOSFET turn-off, when using three-level voltage modulation. However, with the implementation of snubber circuits, it was possible to reduce teh dv/dt slope, as shown in Fig. 13b. A reduction of nearly 50% in the voltage variation rate (dv/dt) of V_{DS} was observed during the turn-off of Q_2 when comparing the original circuit, without a snubber, with the version using the RCD snubber. This reduction significantly contributes to the mitigation of voltage overshoots and associated effects such as electromagnetic interference (EMI) and switching losses. For the other MOSFETs, an effective reduction in this rate was also observed with the use of snubber circuits.

Acknowledgment

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